Computing Fundamentals
EC-110

Lecture # 02

Evolution of the Computer

Evolution of the Computer

- **"Computer generation" is a term related to the evolution and adaptation of technology and computing.**

- reducing the size of processors
- Increasing capacity and speed

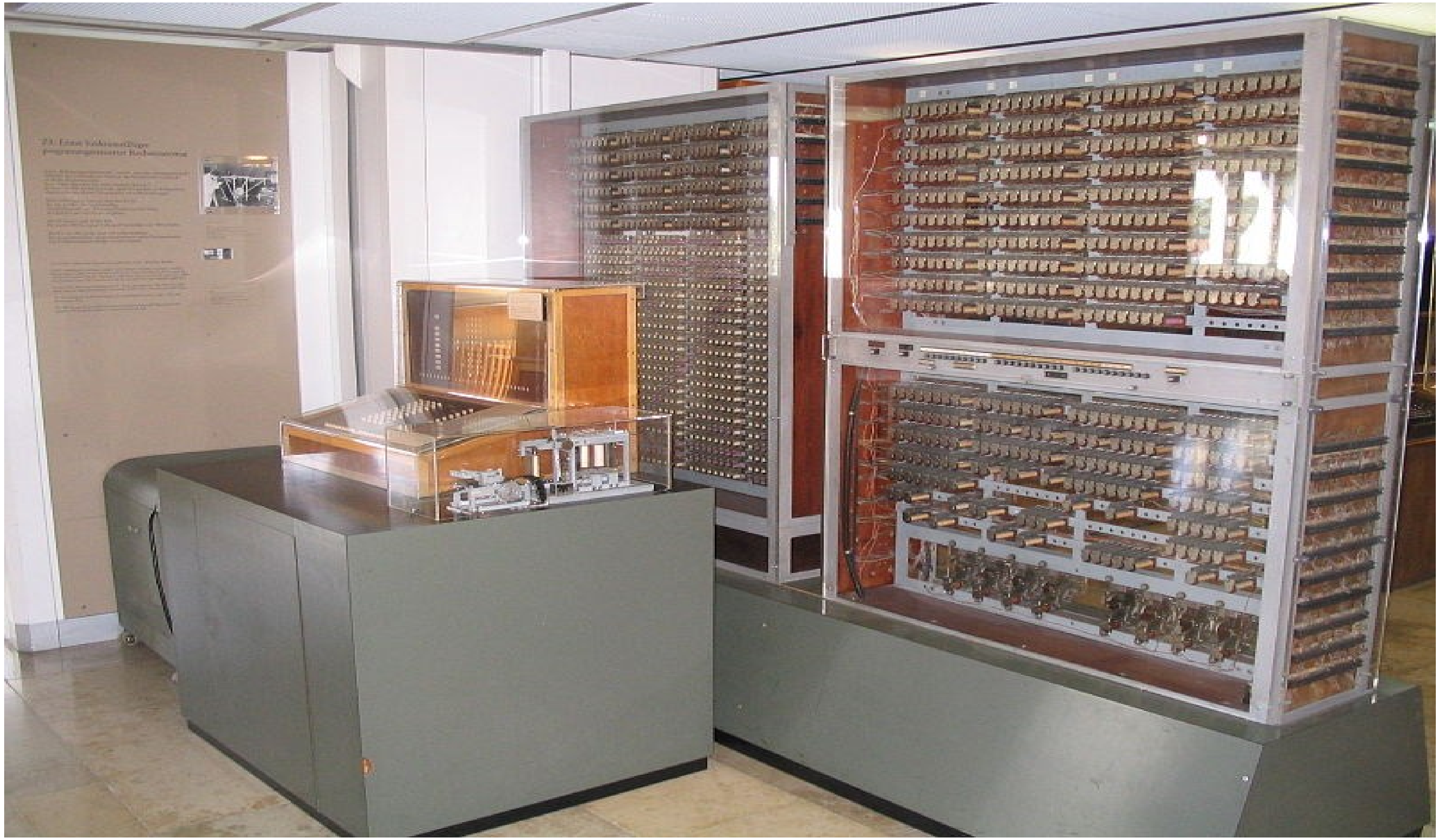
Evolution of the Computer

- First Generation
- Second Generation
- Third Generation
- Fourth Generation
- Fifth Generation

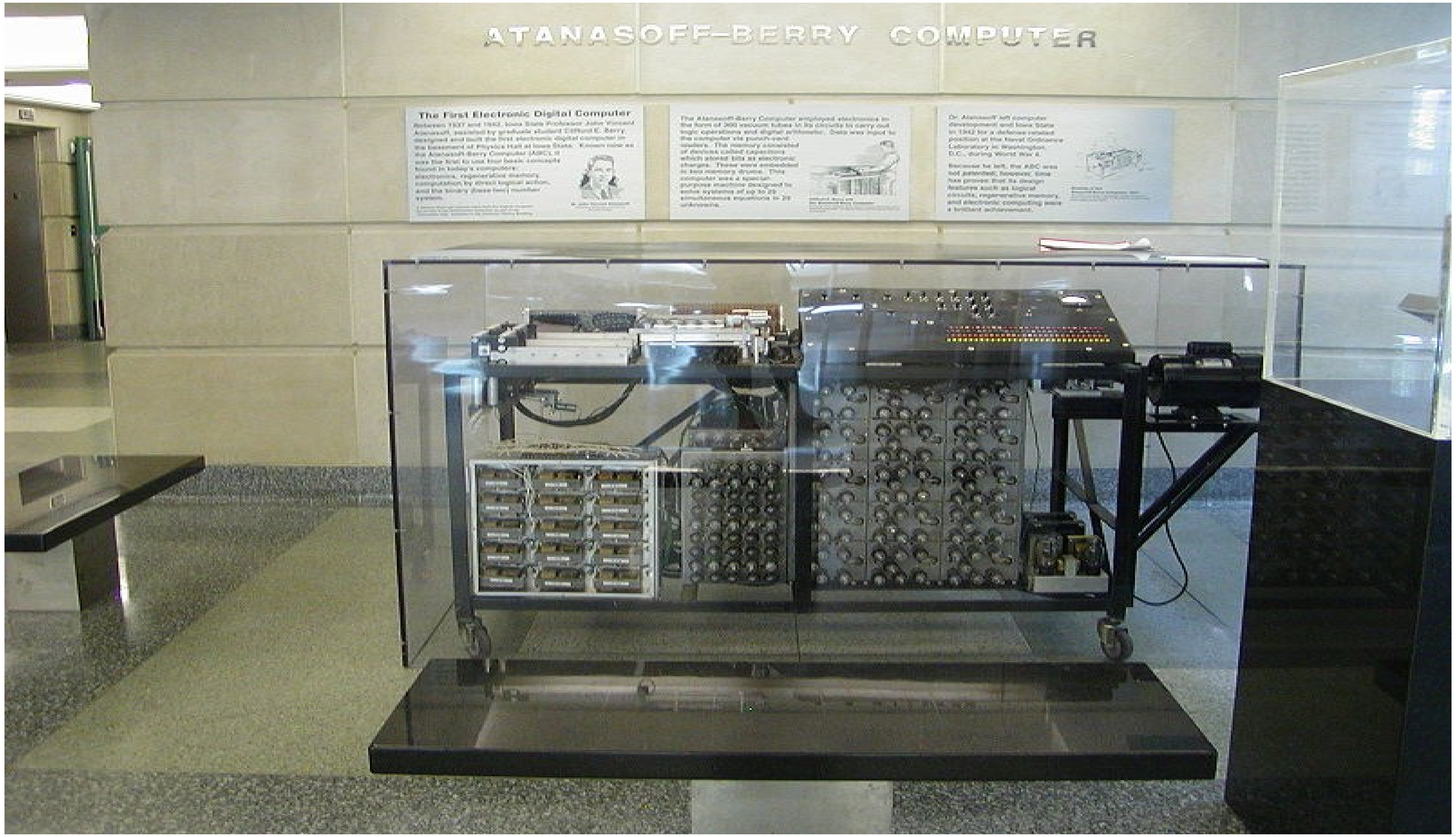
First Generation of Modern Computers (1941-1955)

Date	Scientist	Country	Name	Programmable	Binary	Electronic
May 1941	Conrad Zuse	Germany	Z3	Punched film	Yes	No
Summer 1941	John Atanasoff	Germany/ USA	ABC	No	Yes	Yes
1943	Tommy Flowers	UK	Colossus	By rewiring	Yes	Yes
1944	Howard Aiken/IBM	USA	Harvard Mark I	Punched paper tape	No	Yes
1944	Mauchly & Eckert	USA	ENIAC	By rewiring	No	Yes

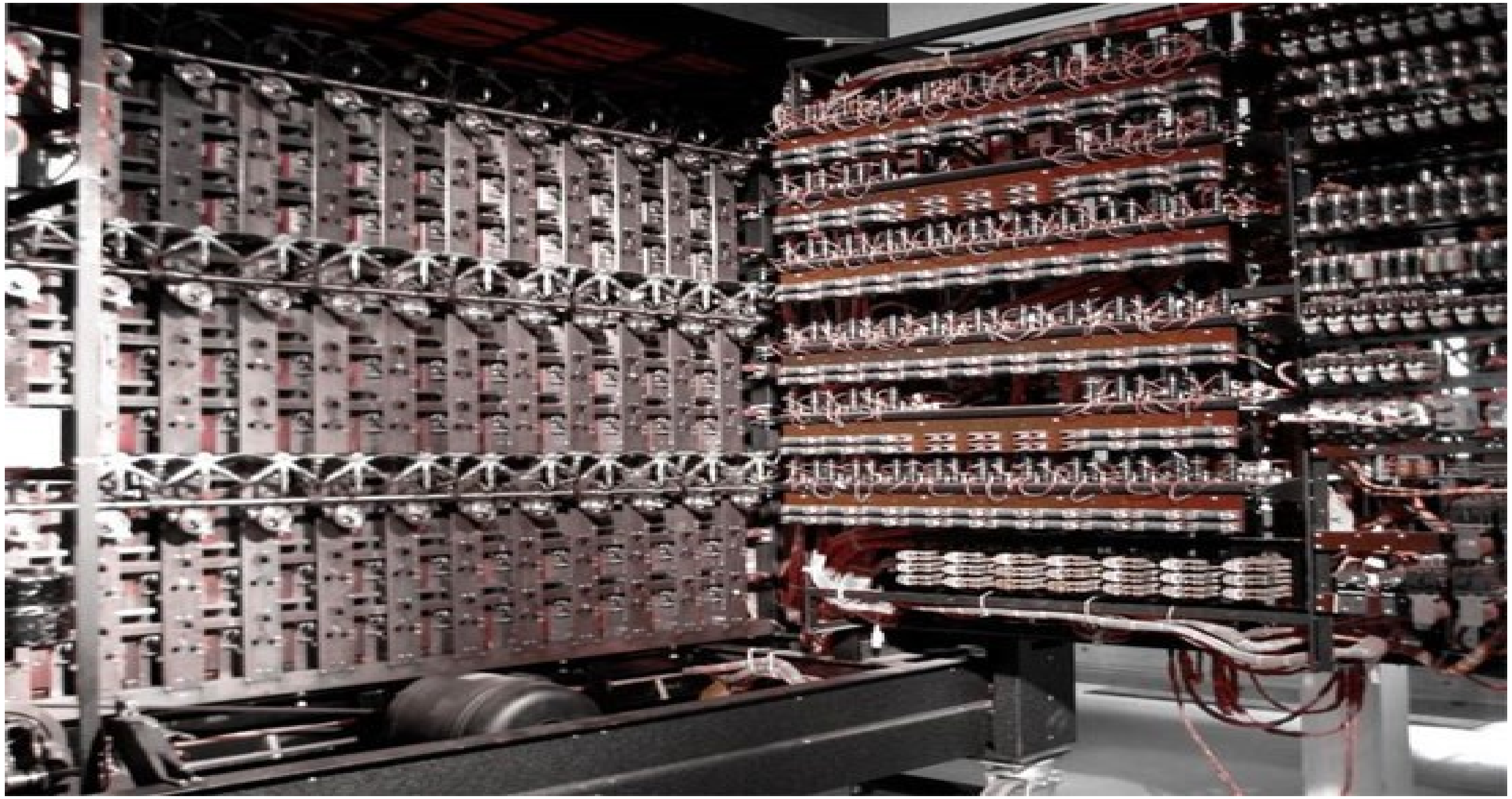
Zuse Z3 in Deutschen Museum in München



Atanasoff–Berry Computer replica at 1st floor of Durham Center, Iowa State University

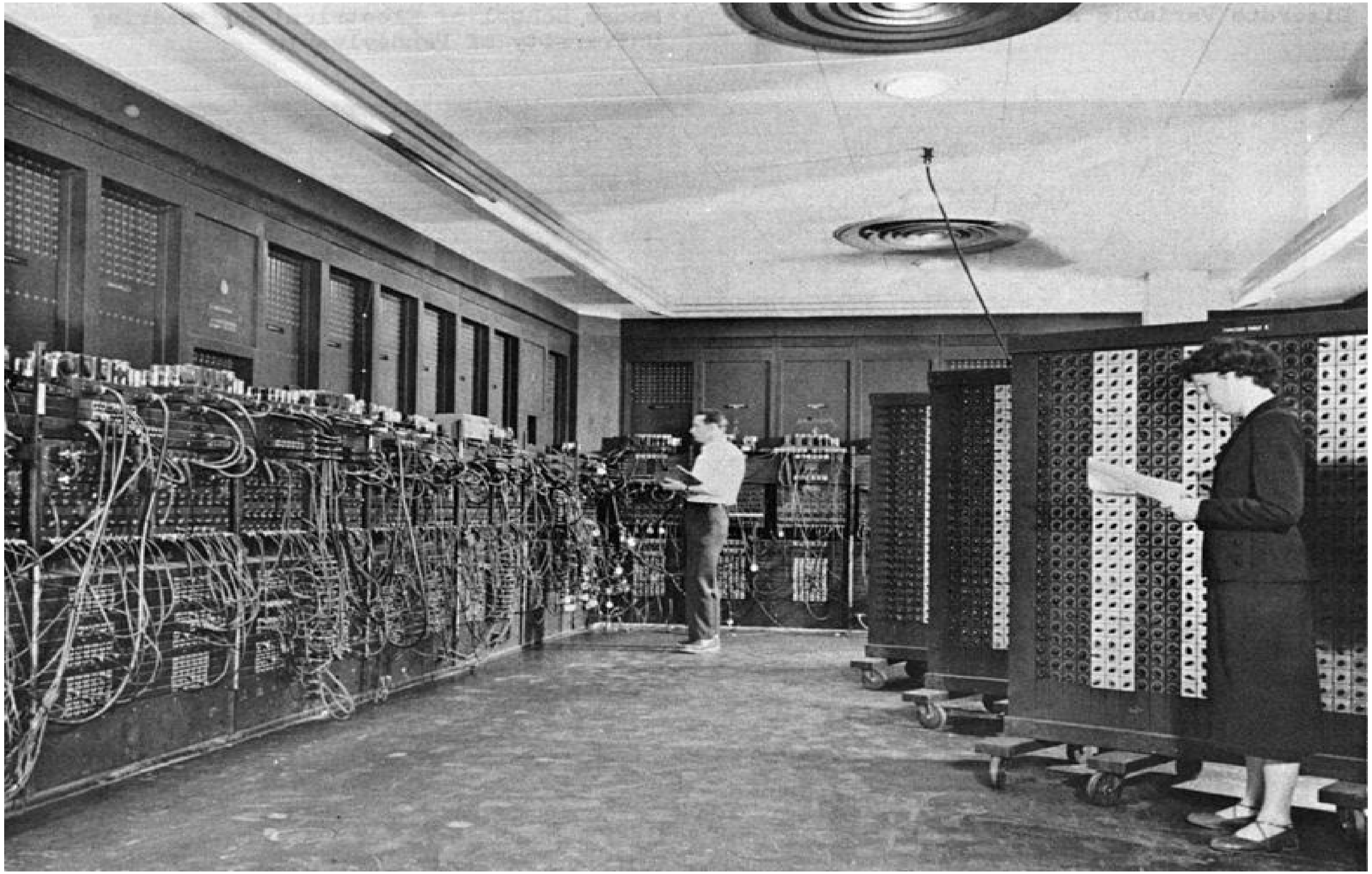


Colossus



Harvard Mark I

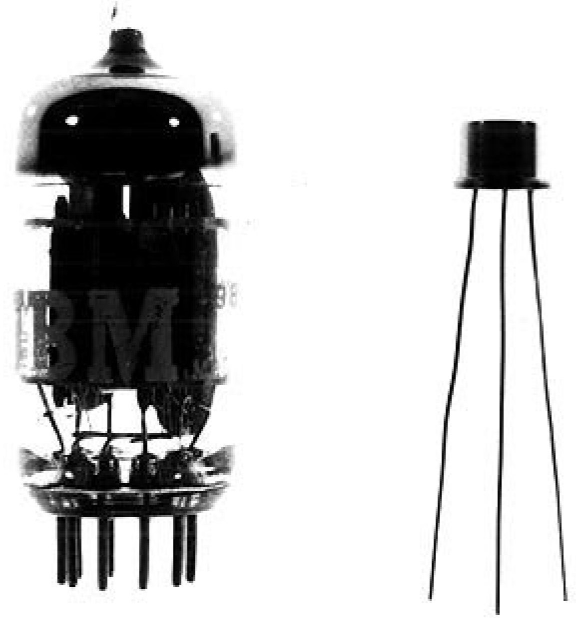
ENIAC



Second-Generation Computers (1956-1963)

- Transistor (1947) replaced vacuum tubes
  - Smaller and less power hungry: larger computers could be built
- Stored-program + Machine Language replaced hard wiring

Vacuum Tube & Transistor



Third Generation Computers (1964-1971)

- Integrated Chips
  - (Many transistors on the integrated on the same chip)
  - Lower Power requirement
  - Faster Computers
- Software
  - Development of high level programming language
  - Easier to program computers
- Machines of this era started making inroads to the business enterprises
- Millions of operations per second

Fourth Generation Computers (1971-1995)

- Very Large Scale Integrated (VLSI) Chips
- Microprocessor
  - Lower Cost
  - Introduction of Personal Computers
- Software
  - Application software, Word processors, gaming etc.
  - AI
- Embedded Systems and Robots
- Hundred million of operations per second or more

Fifth Generation Computers (1995-present)

- Parallel, Grid / Distributed Computing
- Mobile Computing
- Internet/World Wide Web
- Artificial Intelligence
- Software
  - Distributed Object Technologies
  - Middleware
    - Applications are constructed in layers or tiers
    - Maybe distributed along network

Future Trends

- Grid Computing Systems
- Enhancement in Personal/Mobile computing power
- Quantum Computing
  - Using sub-atomic particles to store and manipulate data

Chapter 1B

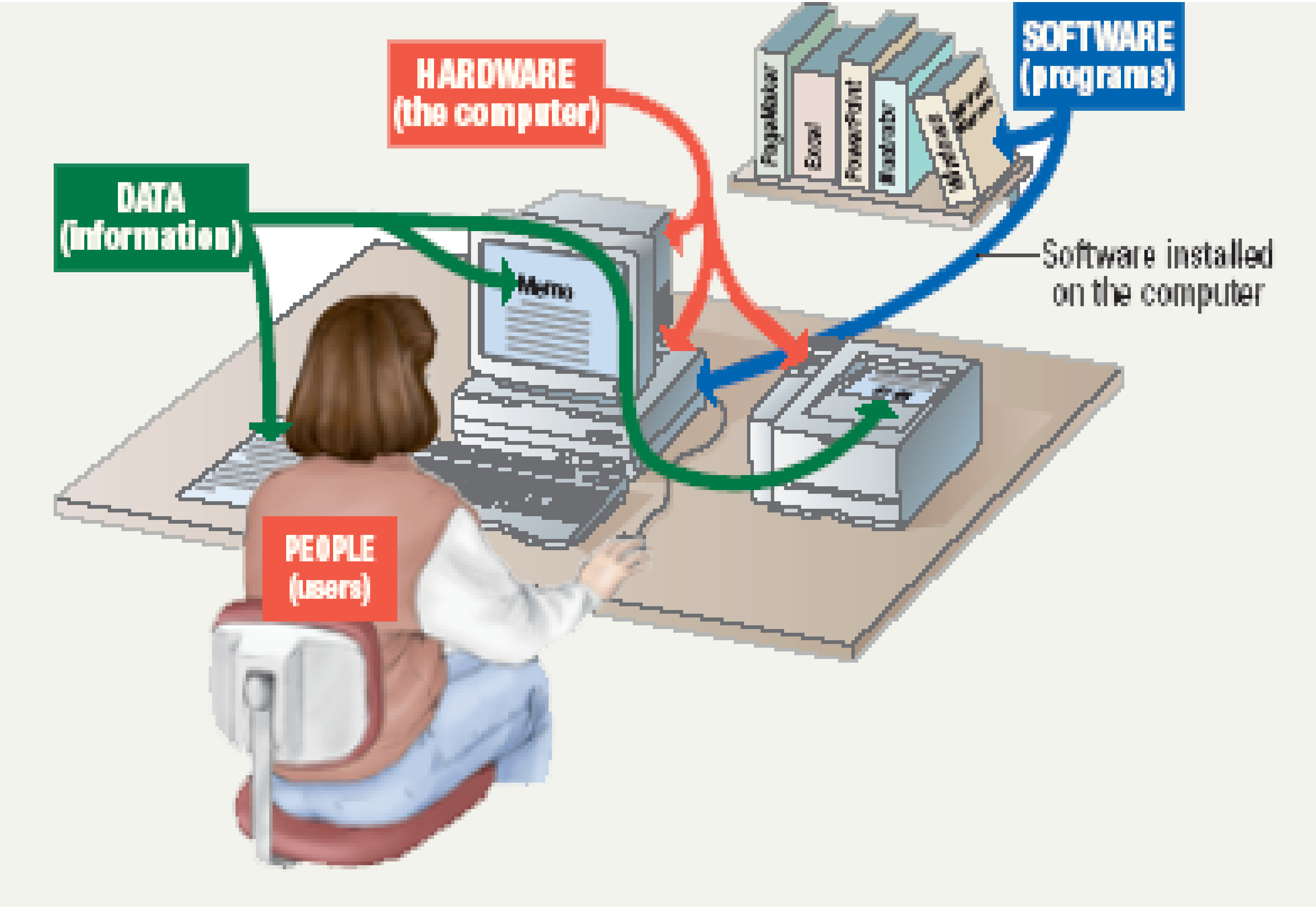
Looking Inside the Computer System

Overview

- Parts of Computer System
  - > Hardware
  - > Software
  - > Data
  - > User
- Information Processing Cycle
- Essential Computer Hardware
  - Processing Devices
  - Memory Devices
    - RAM
    - ROM
  - Input and Output Devices
  - Storage Devices
    - Magnetic Storage
    - Optical Storage

Parts of the Computer System

- Computer systems have four parts
  - Hardware
  - Software
  - Data
  - User
- **Hardware**
  - Mechanical devices in the computer
  - Anything that can be touched
  - Interconnected electronic devices used to control computer's operations, input, output
  - Referred to as device

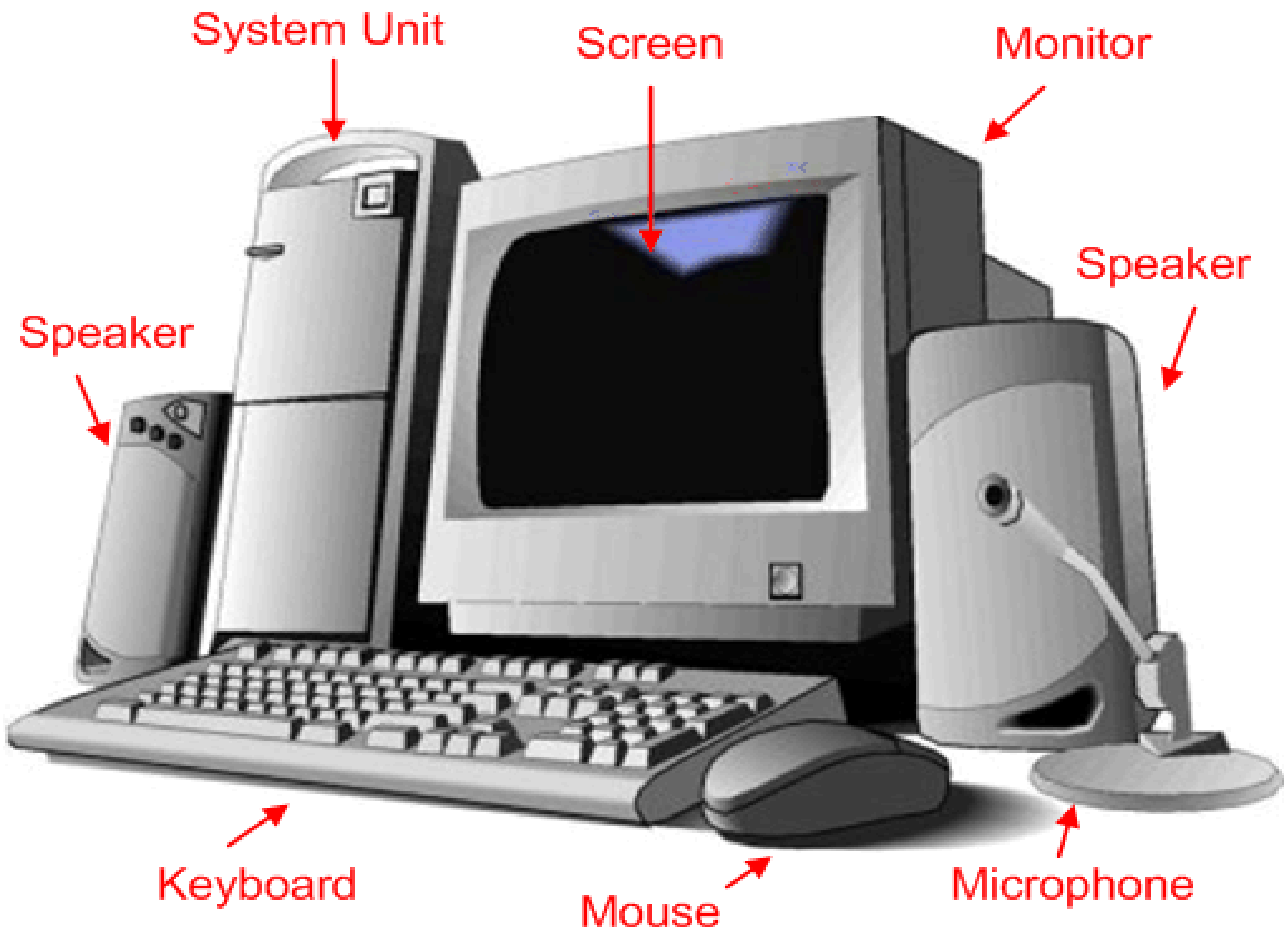

HARDWARE
(the computer)
SOFTWARE
(programs)
DATA
(information)
Memo
Software installed
on the computer
PEOPLE
(users)

Parts of the Computer System

- **Software**
  - Set of instructions which make the computer work
  - Tell the computer what to do
  - Also called a program
  - Thousands of programs exist
  - Some to help computer perform its tasks and manage its resources, others to help users perform their tasks such as creating documents

Parts of the Computer System

- **Data**
  - Pieces of information that by themselves do not make much sense
  - Computer processes them in various ways, converting them to useful information
  - Computer organize and present data
- **Users**
  - People operating the computer
  - Most important part
  - Tell the computer what to do
  - No system is completely autonomous

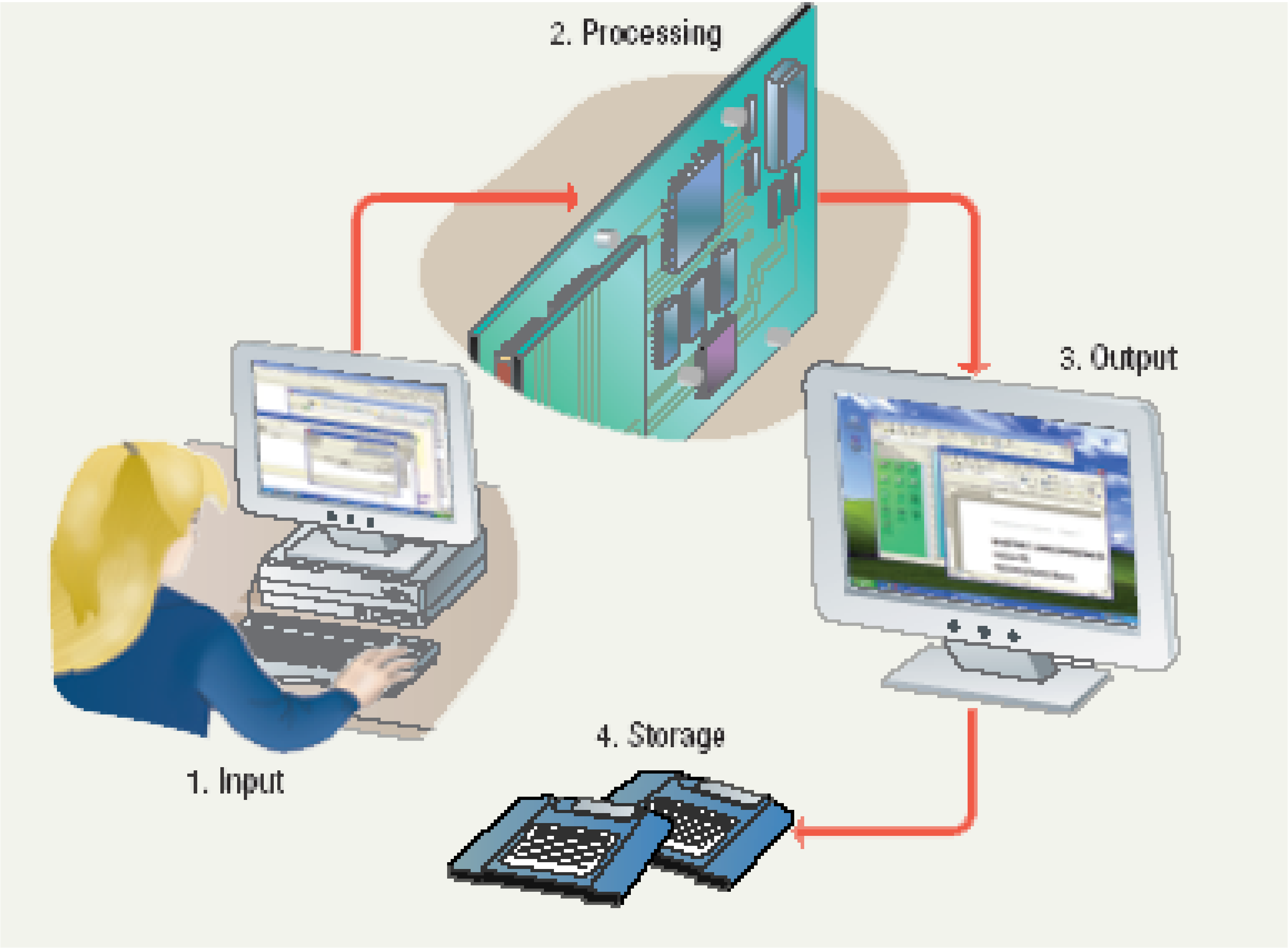

System Unit
Screen
Monitor
Speaker
Speaker
Keyboard
Mouse
Microphone

Information Processing Cycle

Steps followed to process data

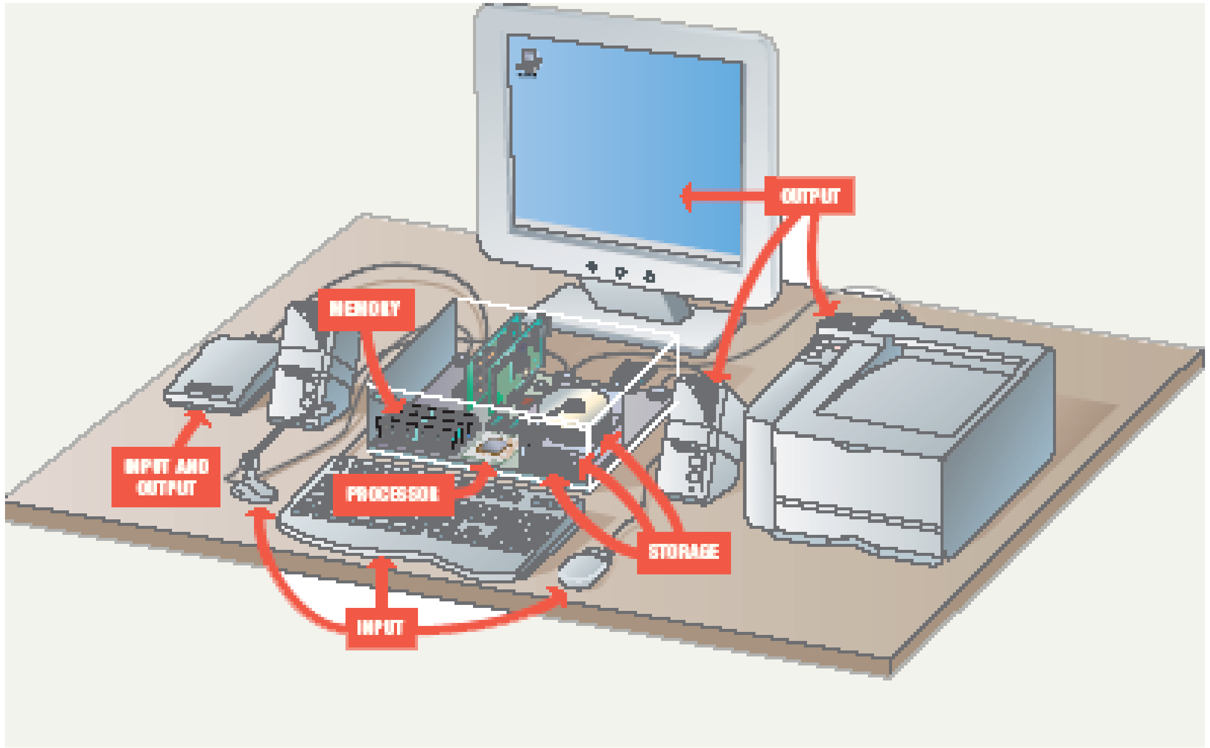
- A computer converts data into information by performing various operations on data according to some instructions from a program, displays results to user and stores them
- This is called *Information Processing Cycle*
- Following are the steps, and each involves one or more specific components of computer
  - **Input**
  - **Processing**
  - **Output**
  - **Storage**

optional


2. Processing
3. Output
1. Input
4. Storage

Essential Computer Hardware

- Computers use the same basic hardware
- Hardware categorized into four types



Essential Computer Hardware

- Processing devices
  - Brains of the computer
  - Carries out instructions from the program
  - Manipulate the data
  - Most computers have several processors
  - Central Processing Unit (CPU)
  - Secondary processors
  - Processors made of silicon and copper

Essential Computer Hardware

- Memory devices
  - Stores data or programs
  - Random Access Memory (RAM)
    - Volatile
    - Stores current data and programs
    - More RAM results in a faster system
  - Read Only Memory (ROM)
    - Permanent storage of programs
    - Holds the computer boot directions

Essential Computer Hardware

- Input and output devices
  - Allows the user to interact
  - Input devices accept data
    - Keyboard, mouse
  - Output devices deliver data
    - Monitor, printer, speaker
  - Some devices are input and output
    - Touch screens

Essential Computer Hardware

- Storage devices
  - Hold data and programs permanently
  - Different from RAM
  - Magnetic storage
    - Floppy and hard drive
    - Uses a magnet to access data
  - Optical storage
    - CD and DVD drives
    - Uses a laser to access data

Software Runs The Machine

- Tells the computer what to do
- Reason people purchase computers
- Two types
  - System software
  - Application software

Software Runs The Machine

- System software
  - Most important software
  - Operating system
    - Windows XP
  - Network operating system (OS)
    - Windows Server 2003
  - Utility
    - Symantec AntiVirus

Software Runs The Machine

- Application software
  - Accomplishes a specific task
  - Most common type of software
    - MS Word
  - Covers most common uses of computers

Computer data

- Fact with no meaning on its own
- Stored using the binary number system
- Data can be organized into files

Computer users

- Role depends on ability
  - Setup the system
  - Install software
  - Mange files
  - Maintain the system
- “Userless” computers
  - Run with no user input
  - Automated systems

Any Questions ?